Abhishek Kumar

AI/ML Engineer

Hyderabad, India | +91-8757479471 | abhibarnwal707@gmail.com | linkedin.com/in/abhibarnwal707

EDUCATION

Indian Institute of Technology (BHU), Varanasi — **Dec 2021 - May 2025**
B.Tech in Chemical Engineering — *CGPA: **8.95/10***

PROFESSIONAL SUMMARY

AI Engineer at Deloitte USI with **1+ year** building production GenAI systems on AWS for Fortune 500 clients: RAG pipelines, multi-agent workflows (LangGraph, PydanticAI) and LLM automation. Skilled in Python, SQL, PyTorch, QLoRA fine-tuning, GraphRAG and classical ML (XGBoost). Contributed to **~$2M** in client savings.

TECHNICAL SKILLS

Languages: Python, SQL, C++
GenAI & Agents: RAG, GraphRAG, Text2Cypher, QLoRA Fine-Tuning, LangGraph, LangChain, PydanticAI, MCP
ML & Data: PyTorch, TensorFlow, XGBoost, FAISS, SHAP, MLflow, Neo4j, MongoDB Vector Search, PostgreSQL, Databricks
Cloud & MLOps: AWS (Bedrock, ECS, Lambda, S3, SQS), FastAPI, Docker, Terraform, Jenkins CI/CD, Streamlit, Git

WORK EXPERIENCE

Deloitte USI — **Jul 2025 - Present**
AI Engineer, AI & Data Practice — *Hyderabad, India*

Agentic Underwriting & Insurance Analytics

- Engineered PydanticAI and LangGraph multi-agent workflows extracting broker-slip clauses, cutting processing from **4 hrs to 10 min**.
- Delivered LLM-driven multi-agent analysis over **2M+** insurance records producing monthly insights for senior stakeholders, cutting manual effort by **80%** and contributing to **~$2M** in program savings.

LLM Data Quality Rules Automation

- Built an LLM pipeline for a Fortune 500 insurer that generates column-level validation rules from metadata, business descriptions and data requirements, cutting manual rule-authoring effort by **60%+**.
- Grounded rule generation with MongoDB Vector Search, stored rules in PostgreSQL, ran compute on Databricks, and added human-in-the-loop approve, reject and refine steps across Medallion (Bronze, Silver, Gold) layers.

AI Document Automation Platform

- Developed **3** FastAPI microservices on AWS ECS; the drafting service generates business documents with RAG on Amazon Bedrock Knowledge Base and a format-preserving DOCX/PDF pipeline.
- Implemented selective LLM section rewriting with full version rollback, and an LLM-as-judge evaluator scoring every document.
- Upgraded RAG retrieval over a **22M-vector** S3 Vectors index with cross-encoder reranking, metadata filters and BM25 + HyDE search.

Imbesideyou Inc. — **May 2024 - Jul 2024**
Data Science Intern — *Remote*

- Fine-tuned BERT for 4-level depression severity classification on **3.5K** posts, reaching **88.5%** test accuracy vs 71.7% for a Random Forest on 2,333 engineered features, with SHAP explanations.
- Designed audio/video pipelines (librosa, MediaPipe) extracting **10** vocal and **7** facial/posture cues to flag deviations from baseline.

PROJECTS

RedString-CTI | *QLoRA, LangGraph, MCP, Neo4j, Text2Cypher* — GitHub | Demo

- Fine-tuned Qwen3-1.7B with QLoRA to extract entities and MITRE ATT&CK techniques from threat reports, beating zero-shot gpt-oss-120b (**70x** larger): entity F1 **0.61 vs 0.32**, technique F1 **0.60 vs 0.45**, **100%** schema-valid JSON.
- Built a **397K-node** Neo4j graph from 5 sources; on a 150-question benchmark, Text2Cypher hit **79.8%** vs 13.8% for vector RAG.
- Developed a LangGraph agent (plan, retrieve, answer, self-check) that queries the graph via a read-only MCP server (**8** tools), with a code-level check dropping any cited ID not in retrieved evidence.

PaperTrail | *XGBoost, PyTorch GNN, LangGraph, FastAPI, Docker* — GitHub | Demo

- Designed an anti-money-laundering system on **5.08M** payments across **515K** accounts (0.1% laundering), blending XGBoost with a PyTorch GINE edge GNN: PR-AUC **0.50** and **72%** recall at 1% false-positive rate.
- Engineered **49** point-in-time account-history features that raised XGBoost PR-AUC **6.7x**; the GNN lifted recall from 66% to **72%** across 8 laundering typologies, with SHAP drivers for every alert.
- Added MLflow, PSI drift monitoring on **26** features with auto-retraining, and a LangGraph agent; served via FastAPI and Docker.

CERTIFICATIONS

AWS Certified AI Practitioner Verify | AWS Certified Cloud Practitioner Verify | Claude Certified Architect - Professional Verify